

Name: _____ **Date:** _____

Guided Lecture Notes

Question 1. Write the Intermediate Value Theorem (IVT).

Question 2. In your own words, what does the Intermediate Value Theorem say?

Question 3. Use the Intermediate Value Theorem to show that the function $f(x) = x^3$ has a zero on the interval $[-1, 1]$.

Question 4. Use the Intermediate Value Theorem to show that the function $f(x) = 5x^5 - x^4 + 3x^3 + x^2 - 2$ has a zero on the interval $[-1, 1]$.

Question 5. Use the Intermediate Value Theorem to show that the function $f(x) = x^2$ and $g(x) = \cos(x)$ intersect somewhere on the interval $[0, \frac{\pi}{2}]$.

Practice Problems

Question 6. Use the Intermediate Value Theorem to show that the function $f(x) = x^3 + 2x^2 - 2$ has a zero on the interval $[-2, 2]$.

Question 7. Use the Intermediate Value Theorem to show that the function $f(x) = e^x - 2$ has a zero on the interval $[0, 1]$.

Question 8. Use the Intermediate Value Theorem to show that the functions $f(x) = x^2$ and $g(x) = \sqrt{x+1}$ intersect on the interval $[-1, 1]$.

Question 9. Use the Intermediate Value Theorem to show that the functions $f(x) = \sin(x)$ and $g(x) = \cos(x)$ intersect on the interval $[0, \frac{\pi}{2}]$.